

Project : Yesterday's Shadow

Yesterday's Shadow is a 3D puzzle, time-manipulation, and stealth game developed using Python, PyOpenGL, GLUT, and GLU.

The player is trapped inside a haunted castle and must escape from a series of sealed rooms. The central challenge is that the player has only one body and cannot press several switches or complete multiple tasks at the same time.

To solve this problem, the player uses a **Time-Freeze Spell**. When the spell is activated, the level timer pauses and a short Magic Window begins. During this window, the player can move and perform an action, such as walking onto a press

The game contains three rooms with increasing difficulty.

The player begins each room with a limited room timer, spell charges, a torch, health, and lives. The objective is to solve the room's Ghost-based puzzle and reach the exit before time runs out.

The player can use the Time-Freeze Spell to record movement. After the recording ends, a Ghost repeats that movement and can remain on a pressure button. The player must plan the placement of Ghosts carefully because later rooms require multiple Ghosts to hold multiple buttons at the same time.

The player must also manage the torch. When the torch is on, coins can be collected and the room is visible, but the Shadow can detect and chase the player in later levels. When the torch is off, the Shadow stops chasing, but the screen becomes dark.

Coins are collected from hidden pickups, required button activations, level completion rewards, and speed bonuses. The player can spend coins in the shop to buy useful upgrades.

Main Features

1. Time-Freeze Spell Casting System
2. Magic Window Timer
3. Ghost Recording System
4. Ghost Replay AI
5. Multi-Ghost Management
6. Simultaneous-Interaction Detection
7. Door and Gate Unlock Logic
8. Torch System
9. Darkness and Limited Visibility
10. Player Movement
11. Checkpoint and Restart System
12. Level Progression
13. Increasing Difficulty every level
14. Coin Collection
15. Shop and Purchase System
16. HUD
17. Interactive Room Objects
18. Sliding Wall System (Level 3)
19. Hold-Down Button and Wall-Halt Logic (Level 3)
20. Spike Hazard Trap
21. Extraction Portal / Win Condition
22. Fail State and Retry System
23. Save / Progress System
24. Chasing Shadow AI
25. Minimap / Room Indicator
26. Ending Sequence

Explanation

Level 1: Introduction to Ghost Puzzles

Level 1 teaches the main Ghost mechanic.

The player must create a Ghost and leave it on Button A. Then the player moves to Button B. When both buttons are pressed at the same time, the eastern exit opens.

This level teaches the player how Time-Freeze, recording, Ghost replay, and simultaneous button activation work together.

Level 2: Two-Ghost Puzzle

Level 2 increases the difficulty by requiring two Ghosts.

The player must create one Ghost for Button A and another Ghost for Button B. Unlike Level 1, the buttons must be held by Ghosts. Once both Ghosts are correctly placed, the exit opens and the player can proceed.

This level teaches planning, timing, and managing multiple Ghost recordings.

Level 3: Final Escape

Level 3 is the final challenge.

The player must place Ghosts on all three buttons to halt the wall gates and open the extraction portal. The player must then reach the portal while managing hazards, the Shadow, torch charge, health, and remaining time.

Completing this level triggers the victory screen and finishes the game.

Controls

- **W, A, S, D / Arrow Keys:** Move the player
- **F:** Use the Time-Freeze Spell and record an action
- **T:** Toggle the torch
- **K:** Open the shop
- **R:** Restart the current level
- **ESC:** Pause the game or return from menus
- **Mouse:** Select menu and shop options

Summary

Our project is called Yesterday's Shadow. It is a third-person puzzle, time-manipulation and stealth game. The player is trapped inside a haunted castle and has to escape one room at a time. The main mechanic is a Time-Freeze Spell. When the player uses it, the room timer pauses and the player gets a short magic window to perform one action. After that, a ghost appears and repeats exactly what the player did, while the main timer starts again. The player can then do another action somewhere else. By using the player and multiple ghosts together, we can solve puzzles where several buttons or pressure points need to be active at the same time.

There are also dark areas where the player needs a torch, coins that can be collected and spent in a shop, upgrades like extra spell charges and extra ghost slots, checkpoints, a HUD, increasing difficulty and different interactive objects. The final level is called The Closing Walls. It has three sliding walls with hold-down buttons. The player uses the Time-Freeze Spell to create ghosts that keep the earlier buttons pressed while the player moves forward. The final level also includes the spike section, Rift tiles and the Shadow chase, and the player has to reach the extraction portal to complete the game.

Features Worked on :

MD. Neamautullah Rahat	MD. Arafat Choudhury	Ali Ahbab
• Time-Freeze Spell Casting System • Magic Window Timer • Ghost Recording System • Ghost Replay AI • Multi-Ghost Management • Simultaneous-Interaction Detection • Door and Gate Unlock Logic • Level Progression	• Torch System • Darkness and Limited Visibility • Player Movement • Checkpoint and Restart System • Interactive Room Objects • Sliding Wall System (Level 3) • Hold-Down Button and Wall-Halt Logic (Level 3) • Fail State and Retry System • Save / Progress System	• Increasing Difficulty every level • Coin Collection • Shop and Purchase System • HUD • Spike Hazard Trap • Extraction Portal / Win Condition • Chasing Shadow AI • Minimap / Room Indicator • Ending Sequence